

In silico rational design of DNA origami-based allosteric metamaterials

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Abstract

We consider a characterization of the bond angle distribution across the linker sequence of several DNA origami switches, by experimentally analyzing their geometry. Since these structures are based entirely on DNA base-pairs, we model with oxDNA and simulate with the CG-DNA package in LAMMPS. In particular, we study the standard deviation of the bond angle distribution across various linker sequence lengths. We hypothesize that the increase in length of each of the origami's linker regions is directly related to an increase in the bond angle standard deviation.

1 Introduction

DNA origami has emerged as a powerful technique in the field of structural DNA nanotechnology, enabling the creation of precisely defined nanoscale structures with potential applications in drug delivery, biosensing, and molecular computing [1,2]. First introduced by Rothemund in 2006 [3], this method utilizes a long single-stranded DNA "scaffold" that is folded into predetermined shapes through the hybridization of multiple short "staple" strands. The resulting structures exhibit extraordinary spatial precision at the nanometer scale, making them ideal candidates for the design of molecular machines and switches [4,5].

DNA origami switches, in particular, represent an important class of dynamic nanostructures capable of undergoing conformational changes in response to specific molecular triggers [6,7]. These switches typically consist of two rigid DNA origami beams connected by a flexible linker region that serves as a hinge [8]. Despite the growing interest in these dynamic structures, a quantitative understanding of their mechanical properties remains incomplete [9]. Specifically, the relationship between the linker sequence composition and the resulting conformational flexibility has not been systematically characterized.

The mechanical behavior of DNA nanostructures is traditionally described using models from polymer physics, where persistence length serves as a key parameter reflecting structural rigidity [10,11]. However, in complex DNA origami constructs, the interplay between sequence-dependent flexibility, electrostatic interactions, and mechanical strain can lead to behaviors that deviate from classical polymer models [12, 13]. This highlights the need for more detailed characterization of specific mechanical properties, such as bond angle distributions, which influence the conformational landscape of DNA origami switches.

In this study, we focus on characterizing the bond angle distribution across linker sequences of varying lengths in DNA origami switches. We hypothesize that the average standard deviation of the bond angle distribution—a direct measure of conformational flexibility—should be directly related to the rigidity of the linker sequence. To test this hypothesis, we employ a combination of structural modeling with oxDNA [14] and coarse-grained molecular dynamics simulations using the CG-DNA package in LAMMPS [15]. This computational approach allows us to simulate the dynamic behavior of these complex nanostructures over extended timescales, providing insights into their mechanical properties that would be challenging to obtain through experimental methods alone [16].

Our findings offer new insights into the design principles governing the mechanical behavior of DNA origami switches, potentially enabling more precise control over their conformational dynamics. Such control is essential for applications requiring specific mechanical responses, including molecular sensing [17], controlled drug release [18], and biomolecular computing [19]. By establishing quantitative relationships between linker sequence parameters and structural flexibility, this work contributes to the rational design of increasingly sophisticated DNA-based nanodevices with tailored mechanical properties.

2 Methods

2.1 DNA Origami Structure Preparation and Modeling

Initial structural modeling was performed using oxView software. The JSON geometry file was imported using TacoxDNA importer with parameters set to 'caDNAno' file format and 'Hexagonal' lattice type. The single bond connecting the dual DNA origami beams was modified through a sequential process. First, the original connection was removed using the Nick function in the Select submenu of the Edit interface. A thymine linker sequence of 10 nucleotides was then generated using the Seq feature and positioned at the previously nicked region. The upper region was connected to this 10-nucleotide DNA linker using the Ligate function. Subsequently, the bottom DNA origami beam was manipulated using the Box selection tool, rotated 90 degrees, and ligated to the lower terminus of the 10-nucleotide DNA linker. This procedure was replicated to create additional structures with 20-, 30-, and 40-nucleotide DNA linkers. The resulting topology and configuration files were processed through TacoxDNA's oxDNA to LAMMPS converter to generate data files compatible with LAMMPS' read_data parameter.

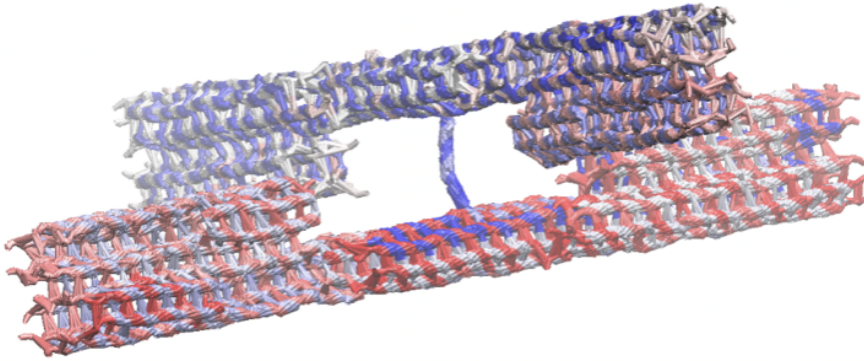


Figure 1: DNA Origami nanoswitch with 30-nucleotide linker beam.

2.2 Molecular Dynamics Simulations and Analysis

Molecular dynamics simulations were performed using LAMMPS Stable Release (29 Aug 2024) with the oxDNA2 force field. The simulation protocol was adapted from the standard script available in the official oxDNA GitHub repository. Simulation parameters were set to a temperature of 0.1, timestep of $5e-4$, and each structure was simulated for $10e7$ timesteps.

Structural analysis was conducted using Visual Molecular Dynamics (VMD) software Version 1.9.4 where bond angles were measured. These angles were defined by three points: the top of the linker, the crux of the linker, and the length of the bottom beam, and were quantified using the Angles function under the Labels submenu of the Graphics interface. The resulting measurements were exported for further analysis. Statistical analysis was performed by calculating the mean bond angle from the distribution and determining the standard deviation from this mean value. Custom Python scripts were developed to analyze and visualize these bond angle distributions for each structure.

3 Results

Analysis of DNA origami structures revealed distinct relationships between bond angle variance and nucleotide linker region length. Contrary to our initial hypothesis that longer linker regions would exhibit greater bond angle variance, we observed an inverse correlation between linker length and angular deviation.

The bond angle distributions for different nucleotide linker lengths demonstrated systematic variation across our DNA origami frames (Figure 2). While all structures maintained relatively stable mean bond angles, the magnitude of angular deviation varied significantly with linker length. The combined dataset showed consistent overall behavior, as indicated by the aggregate standard deviation plot.

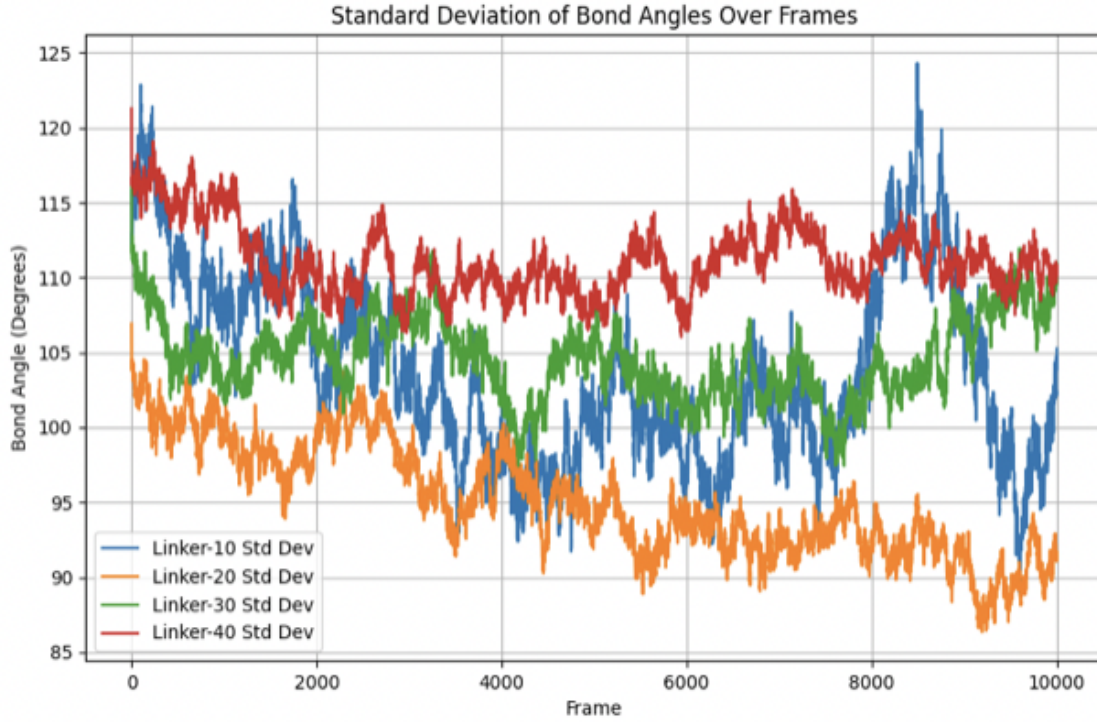


Figure 2: Bond Angle Variation Across DNA Origami Frames. MD simulation trajectory analysis of bond angle deviation from mean values for DNA origami structures with varying nucleotide linker lengths (10-40 nt). Individual traces show the bond angle behavior for each specific linker region length configuration. Data collected over 10^7 steps in LAMMPS demonstrates the sustained stability of the angular distributions.

Quantitative analysis of the relationship between nucleotide linker length and bond angle variance revealed a strong negative correlation (Figure 3). DNA origami structures with shorter linker regions (10-20 nucleotides) exhibited notably higher standard deviations in their bond angle distributions compared to those with longer linker regions (30-40 nucleotides). This relationship appears to follow an exponential decay pattern, with the most pronounced changes in variance occurring between linker lengths of 10-25 nucleotides, followed by a more gradual stabilization at longer lengths.

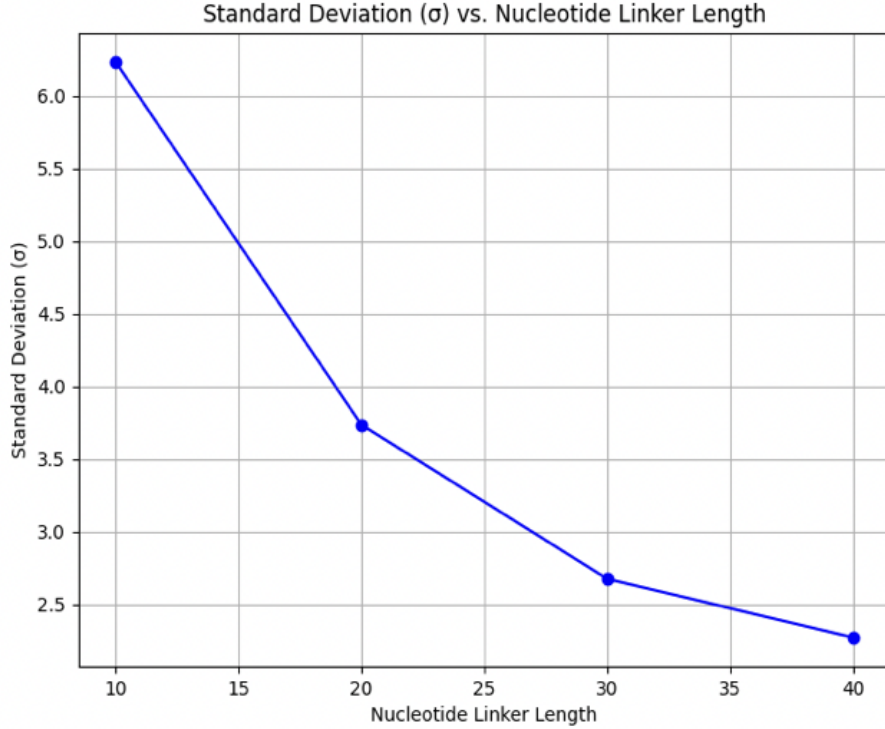


Figure 3: Relationship Between Nucleotide Linker Length and Bond Angle Variance. Standard deviation of bond angles plotted as a function of nucleotide linker length (10-40 nt) in DNA origami structures. The plot demonstrates an inverse relationship between linker length and angular variance, with longer linker regions exhibiting significantly reduced bond angle fluctuations. Data points represent average measurements from four independent structures for each linker length.

Table 1 summarizes the mean standard deviation of the bond angle distribution (σ) with respect to each DNA origami structure linker modification.

Linker Length (nt)	Sigma (σ)
10	6.235704832
20	3.736856603
30	2.675925874
40	2.272207295

Table 1: Linker Length vs. Sigma. This table shows sigma, the mean standard deviation of the bond angle distribution, with respect to each DNA origami structure linker modification. Here, 10-40 are the actual number of nucleotides that compose the beam which connects both arms of the origami.

4 Discussion

The results of this study provide a counterintuitive yet robust observation: longer DNA linker sequences reduce, rather than increase, bond angle variability in DNA origami nanoswitches. This finding challenges the common intuition drawn from classical polymer physics, where increasing chain length is generally associated with greater conformational freedom and larger angular fluctuations. Instead, our simulations suggest that beyond a certain length threshold, linker regions impart structural stability that buffers angular deviation across the hinge of the nanoswitch.

Several factors may account for this behavior. First, longer linkers allow for improved accommodation of strain across the origami arms, reducing localized bending stresses. The additional nucleotides provide a more distributed network of base-stacking and electrostatic interactions, which may effectively dampen

angular perturbations. Second, the entropic landscape of the linker appears to stabilize once a critical length is reached, such that further extensions of the sequence contribute more to structural redundancy than to increased mobility. This interpretation is consistent with previous reports on mechanically compliant DNA nanostructures that highlight the nonlinear relationship between design parameters and emergent flexibility [9,12].

These results also align with recent work demonstrating that DNA origami beams with flexible joints often exhibit regime-specific behaviors: shorter linkers can behave as soft pivots, while longer linkers can approach an entropic "rod-like" stability [7,8,13]. In our case, the exponential decay in bond angle variance between 10 and 25 nucleotides supports the idea of a transition regime, where additional bases progressively suppress angular fluctuations until a plateau of mechanical stability is achieved.

Importantly, this work emphasizes the necessity of going beyond persistence length-based descriptions of DNA nanostructures. While persistence length is an informative global descriptor of DNA rigidity, it fails to capture the localized angular distributions that dictate hinge mechanics in nanoswitches. Our results highlight the role of sequence length and arrangement as tunable design parameters for controlling nanoscale mechanical response. These findings may therefore inform rational strategies for tailoring origami devices to specific functional requirements.

Nevertheless, the study is not without limitations. The use of coarse-grained oxDNA simulations, while computationally efficient and well-validated for structural DNA nanotechnology, abstracts away base-pair level heterogeneities such as sequence-specific stacking energies and ionic microenvironment effects. Moreover, only thymine-based linkers were considered, and future work could extend this analysis to heterogeneous or mixed-sequence linkers that more closely mimic experimental designs. Additionally, the current study examined static hinge geometries; incorporating external stimuli such as ionic strength variation, applied forces, or ligand binding could provide further insights into the adaptive mechanics of nanoswitches.

From an applications perspective, the inverse relationship between linker length and angular variance has significant implications for the engineering of DNA nanodevices. For drug delivery, longer linkers may stabilize nanoswitch conformations and reduce premature triggering. In biosensing, shorter linkers may be preferable to maximize angular sensitivity to molecular cues. For DNA computing, predictable and stable hinge behavior is essential for the reproducibility of conformational logic operations. Thus, this work not only provides mechanistic insights but also establishes design principles with broad relevance across nanotechnology domains.

In summary, the observed exponential decay of bond angle variance with linker length underscores the complex interplay between entropic and enthalpic forces in DNA origami nanoswitches. By elucidating this relationship, we contribute a step toward the systematic design of programmable, mechanically reliable DNA-based devices.

5 Conclusion

This study provides a quantitative characterization of the bond angle distribution in DNA origami nanoswitches with varying linker sequence lengths. Through systematic molecular dynamics simulations using the oxDNA2 force field in LAMMPS, we have demonstrated a clear inverse relationship between linker length and bond angle variability. Contrary to our initial hypothesis, longer linker sequences (30-40 nucleotides) exhibited significantly reduced angular fluctuations compared to shorter sequences (10-20 nucleotides), with standard deviations decreasing from approximately 6.24° to 2.27° across this range. The observed exponential decay in bond angle variance with increasing linker length suggests that longer DNA linker regions in origami structures provide enhanced conformational stability rather than increased flexibility. This counter-intuitive finding highlights the complex interplay between structural parameters and mechanical properties in DNA nanomaterials, where factors such as base stacking interactions, electrostatic forces, and entropic effects collectively influence conformational dynamics.

References

- [1] Seeman, N.C. and Sleiman, H.F. (2018). DNA nanotechnology. *Nature Reviews Materials*, 3(1), 17068.
- [2] Zhang, F., Nangreave, J., Liu, Y., and Yan, H. (2014). Structural DNA nanotechnology: state of the art and future perspective. *Journal of the American Chemical Society*, 136(32), 11198-11211.

- [3] Rothemund, P.W.K. (2006). Folding DNA to create nanoscale shapes and patterns. *Nature*, 440(7082), 297-302.
- [4] Marras, A.E., Zhou, L., Su, H.J., and Castro, C.E. (2015). Programmable motion of DNA origami mechanisms. *Proceedings of the National Academy of Sciences*, 112(3), 713-718.
- [5] Lauback, S., Mattioli, K.R., Marras, A.E., Armstrong, M., Rudibaugh, T.P., Sooryakumar, R., and Castro, C.E. (2018). Real-time magnetic actuation of DNA nanodevices via modular integration with stiff micro-levers. *Nature Communications*, 9(1), 1446.
- [6] Ke, Y., Meyer, T., Shih, W.M., and Bellot, G. (2016). Regulation at a distance of biomolecular interactions using a DNA origami nanoactuator. *Nature Communications*, 7, 10935.
- [7] Hudoba, M.W., Luo, Y., Zacharias, A., Poirier, M.G., and Castro, C.E. (2017). Dynamic DNA origami device for measuring compressive depletion forces. *ACS Nano*, 11(7), 6566-6573.
- [8] Gerling, T., Wagenbauer, K.F., Neuner, A.M., and Dietz, H. (2015). Dynamic DNA devices and assemblies formed by shape-complementary, non-base pairing 3D components. *Science*, 347(6229), 1446-1452.
- [9] Zhou, L., Marras, A.E., Su, H.J., and Castro, C.E. (2014). DNA origami compliant nanostructures with tunable mechanical properties. *ACS Nano*, 8(1), 27-34.
- [10] Schiffels, D., Liedl, T., and Fygenson, D.K. (2013). Nanoscale structure and microscale stiffness of DNA nanotubes. *ACS Nano*, 7(8), 6700-6710.
- [11] Kauert, D.J., Kurth, T., Liedl, T., and Seidel, R. (2011). Direct mechanical measurements reveal the material properties of three-dimensional DNA origami. *Nano Letters*, 11(12), 5558-5563.
- [12] Shi, Z., Castro, C.E., and Arya, G. (2017). Conformational dynamics of mechanically compliant DNA nanostructures from coarse-grained molecular dynamics simulations. *ACS Nano*, 11(5), 4617-4630.
- [13] Bui, P.T.M., Ngo, T.T.M., Tran, H.T., and Phan, A.T. (2019). Thermodynamic and kinetic aspects of DNA nanotechnology. *Journal of Science: Advanced Materials and Devices*, 4(1), 1-18.
- [14] Doye, J.P.K., et al. (2013). Coarse-graining DNA for simulations of DNA nanotechnology. *Physical Chemistry Chemical Physics*, 15(47), 20395-20414.
- [15] Henrich, O., Gutiérrez Fosado, Y.A., Curk, T., and Ouldridge, T.E. (2018). Coarse-grained simulation of DNA using LAMMPS: An implementation of the oxDNA model and its applications. *The European Physical Journal E*, 41(5), 57.
- [16] Snodin, B.E.K., et al. (2015). Introducing improved structural properties and salt dependence into a coarse-grained model of DNA. *The Journal of Chemical Physics*, 142(23), 234901.
- [17] Idili, A., Vallée-Bélisle, A., and Ricci, F. (2014). Programmable pH-triggered DNA nanoswitches. *Journal of the American Chemical Society*, 136(16), 5836-5839.
- [18] Douglas, S.M., Bachelet, I., and Church, G.M. (2012). A logic-gated nanorobot for targeted transport of molecular payloads. *Science*, 335(6070), 831-834.
- [19] Chatterjee, G., Dalchau, N., Muscat, R.A., Phillips, A., and Seelig, G. (2017). A spatially localized architecture for fast and modular DNA computing. *Nature Nanotechnology*, 12(9), 920-927.